

Hotel Booking Demand & Operations Analysis Report

Executive Report | Dataset Scope: 119,390 Historical Records (City & Resort Hotels)

Executive Summary: This report provides an in-depth analysis of 119,390 hotel booking records across City and Resort Hotels. It covers operational efficiency, cancellation trends, revenue dynamics, guest demographics, and channel distribution. The primary findings indicate an overall **cancellation rate of 37.0%**, with City Hotels driving higher volume but suffering greater revenue leakage.

1. Executive Metrics Dashboard

Metric	Resort Hotel	City Hotel	Aggregate Total / Average
Total Volume	40,060 bookings (33.6%)	79,330 bookings (66.4%)	119,390 bookings
Cancellation Rate	27.8%	41.7%	37.0% overall
Average Lead Time	92 days	110 days	104 days
Average Daily Rate (ADR)	€94.95	€105.30	€101.83
Average Length of Stay	4.3 nights	3.0 nights	3.4 nights
Repeat Guest Ratio	4.4%	2.6%	3.2%

2. Cancellation Dynamics & Risk Factors

Lead Time vs. Cancellation Risk Breakdown:

- **0 - 14 Days:** ~18% Cancellation Rate
- **15 - 60 Days:** ~32% Cancellation Rate
- **61 - 120 Days:** ~45% Cancellation Rate
- **120+ Days:** ~54% Cancellation Rate

Key Risk Drivers:

- **Deposit Type Impact:** "No Deposit" bookings show a **36.7%** cancellation rate. "Non-Refund" deposits have near-zero cancellations but are underutilized.
- **Booking Changes:** Guests making ≥ 1 alteration have significantly lower cancellation rates (**14.2%**) vs zero changes (**41.1%**).
- **Previous Cancellations:** Guests with prior cancellations carry an **82.5%** probability of cancelling again.

3. Revenue & Pricing Analysis (ADR & Distribution)

Month-by-Month Average Daily Rate (€):

Month	Resort Hotel	City Hotel
Jan	€48	€82
Apr	€62	€111
Jul	€150	€115
Aug	€181	€118
Oct	€62	€102
Dec	€68	€88

Distribution Channel Performance:

Distribution Channel	Booking Share	Cancellation Rate	Average ADR	Dominant Market
Online Travel Agents (OTA)	47.1%	41.1%	€116.50	Individual Leisure / Business
Offline Travel Agents / TA	20.3%	34.3%	€87.20	Group Bookings / Tour Packages
Direct	10.6%	15.3%	€106.10	Brand Loyalists / Walk-ins
Corporate	4.4%	18.7%	€69.30	Business / Relocation
Groups	16.6%	46.1%	€78.40	Events / Conferences

4. Guest Demographics & Operational Logistics

Top 10 Origin Countries: Portugal (PRT): 40,328 (33.8%), Great Britain (GBR): 12,129 (10.2%), France (FRA): 10,415 (8.7%), Spain (ESP): 8,568 (7.2%), Germany (DEU): 7,287 (6.1%), Italy (ITA): 3,766 (3.2%), Ireland (IRL): 3,375 (2.8%), Belgium (BEL): 2,342 (2.0%), Brazil (BRA): 2,224 (1.9%), Netherlands (NLD): 2,104 (1.8%).

Meal Plan Preferences: Bed & Breakfast (BB): 77.3%, Half Board (HB): 12.1%, Self Catering (SC): 8.9%, Full Board (FB): 0.7%.

Room Assignment & Upgrades: 10.3% of bookings received a different room type than reserved (typically upgraded due to inventory constraints), resulting in a cancellation rate of under 5% for affected

5. Strategic Recommendations

- 1. Overbooking Strategy for High-Risk Windows:** Implement a dynamic overbooking factor (10–15%) for OTA bookings made >90 days in advance to offset the 50%+ cancellation rate in that bucket.
- 2. Dynamic Deposit Rules:** Enforce non-refundable 1-night deposit requirements for stays booked during peak summer months (July–August) and long lead times (>60 days).
- 3. Resort Off-Peak Monetization:** Address the steep Resort ADR drops (€181 in Aug vs. €48 in Jan) by targeting long-stay remote workers, wellness retreats, and regional conference bookings in Q1 and Q4.
- 4. Direct Channel Growth:** Offer immediate perks (e.g., free breakfast upgrades or late checkout) for direct bookings to reduce reliance on high-cancellation, high-commission OTA channels.

6. Jupyter Notebook Output Results & Explanation

In[12]: Descriptive Statistics Execution Output

```
import pandas as pd
df = pd.read_csv("hotel_bookings.csv")
print(df.groupby('hotel')[['lead_time', 'adr', 'is_canceled']].mean().round(2))
```

```
lead_time  adr  is_canceled  hotel
City Hotel  109.73
105.30  0.42  Resort Hotel  92.67  94.95  0.28
```

Explanation: The execution output confirms a clear structural divide between properties. City Hotel demonstrates longer advance planning cycles (+17 days) and a significantly higher cancellation

In[18]: Correlation Matrix Execution Output

```
corr = df[['is_canceled', 'lead_time',  
'previous_cancellations', 'booking_changes',  
'total_of_special_requests']].corr()  
print(corr['is_canceled'].sort_values(ascending=False))
```

```
is_canceled 1.000000 lead_time 0.293123  
previous_cancellations 0.110133 booking_changes -0.144381  
total_of_special_requests -0.234658 Name: is_canceled,  
dtype: float64
```

Explanation: `lead\_time` and `previous\_cancellations` show positive linear correlation with cancellations, while engagement indicators (`booking\_changes` and `total\_of\_special\_requests`) exhibit negative correlation, marking them as key retention signals.

7. Summary

This study analyzed 119,390 hotel reservations across City and Resort properties between 2015 and 2017. The macroeconomic view reveals a robust aggregate demand offset by a severe 37% cancellation leak. Overall performance is dominated by Online Travel Agencies (47.1% share), short weekend stays for City Hotels, and strong summer seasonality for Resort properties. Revenue optimizations must balance yield adjustments with cancellation hedging mechanisms.

8. Top Hotel Analysis

City Hotel Deep-Dive: Primary use case is corporate travel and short urban stays (average 3.0 nights). It leads volume accounting for

9. Key Findings

Operational Insights:

- **Lead Time Hazard:** Bookings made >120 days prior have a 54% cancellation probability.
- **Deposit Illusion:** Non-refundable deposits represent <12% of total bookings but achieve near 0% churn.
- **Customer Engagement:** Guests making special requests or modifying itineraries are 65% less likely to cancel.

Revenue Insights:

- **OTA Reliance:** OTAs generate 47.1% of volume but carry high commission fees and a 41.1% cancellation rate.
- **Summer Peak Dominance:** July and August generate over 35% of annual Resort property revenue.
- **Loyalty Deficit:** Aggregate repeat guest rate is under 3.5%, demonstrating reliance on costly acquisition.

10. Conclusion

The hotel booking demand dataset illustrates that while volume and daily rates remain healthy, operational efficiency is heavily impacted by unhedged booking lead times and reliance on indirect distribution channels. By modernizing deposit frameworks, dynamically managing overbooking allowances based on lead time thresholds, and offering direct booking incentives to drive repeat visits, both properties can convert lost capacity into reliable, high-margin revenue.

11. Project Artifacts

Artifact Name	Type	Description
		Raw dataset